

```
$ let x=y=8 z='3 << 4 | x | 2 << y' w=z/3
$ echo $w
109
$ declare -i x=y=8 z='3 << 4 | x | 2 << y' w=z/3
$ echo $w
109
```

Note that the `$(...)` has been replaced by single quoting in assigning the value to `z`. In fact, in most cases—see the assignment of `w`—it can be actually assigned directly as in `C`. Though the single quote for `z` is required to prevent the shell from specially interpreting constructs like `<<`, `>>`, `|`, `*`, etc, and for it not to be space-sensitive. Otherwise, checkout this example:

```
$ declare -i x=y=8 z+=x+y
$ echo $z
17
```

In fact, the variables can be declared as integer once, and then be operated any number of times, as follows:

```
$ declare -i x y z
$ x=y=8
$ z+=x+y
$ echo $z
```

17

And, finally, here are two more illustrative examples:

```
$ echo $(2 ** 100)
0
$ echo $(2.6 + 3)
bash: 2.6 + 3: syntax error: invalid arithmetic operator (error token is ".6 + 3")
```

Oops! How come 2^{100} is zero, and we are not able to do real number operations? It is limited by the C's long integer math, typically upto 64-bit computations on today's computers; and by not having support for non-integer math. That's when we would need to look for something more powerful: the *bench calculator* – which we can explore later! 

By: Anil Kumar Pugalía

The author is a hobbyist in open source hardware and software, with a passion for math. His exploration with math in every aspect of life dates back to the 1990s. A gold medalist from the Indian Institute of Science, math and knowledge-sharing are two of his many passions. Apart from that, he experiments with Linux and embedded systems to share his learning through his weekend workshops. Learn more about him and his experiments at <http://sysplay.in>. He can be reached at email@sarika-pugs.com.

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